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Glossary

Acetylecholine – A neurotransmitter found throughout the nervous system.

Actin – A protein found in the cytoskeleton and muscle cells; it is the principal constituent of the thin filament.

Adenosine triphosphate(ATP) – A nucleotide molecule consisting of adenine, riboseand three phosphate molecules. It plays a central role in energy exchange in biological systems.

Cartilage – A firm, elastic connective tissue produced by the cells,called chondrocytes.

Exoskeleton – Skeletal elements are located upon body surface or in the skin (Example: Shells of snails in inverbrates, Hair claw and nails in vertibrates.

Endoskeleton – Skeletal elements are located inside the organisms with muscles outside. Found in skeletal system of vertibrates

Lever system – Movement takes place along the joints which act as fulcrum of the lever. One can observe functioning of all the three types of levers in the human skeleton.

Mesoderm – The middle embryonic germ layer. It gives rise to the muscular,skeletal,urogenital and circulatory system.

Motor neuron – A motor neuron that transmits nervous impulses from the spinal cord to effectors.

Myoglobin – Heme containing protein that binds molecular oxygen in muscle cells.

Myosin – A protein found in muscle cells that function in muscle contaction. It is present in thick filaments of muscles, known as myosin fibres

Sarcolemma – Muscle cell membrane capable of propagating action potentials

Sarcomere – The functional contractile unit of striated muscle.

Sarcoplasmic reticulum – The endoplasmic reticulum of a muscle cell. It envelopes myofibrils

Tendon – A fibrous connective tissue that connects a bone to a muscle

Evaluation

1. Muscles are derived from
   a. ectoderm b. mesoderm
   c. endoderm d. neuro ectoderm
2. Muscles are formed by
   a. myocytes b. leucocytes
   c. osteocytes d. lymphocytes
3. The muscles attached to the bones are called
   a. skeletal muscle
   b. cardiac muscle
   c. involuntary muscle
   d. smooth muscles
4. Skeletal muscles are attached to the bones by
   a. tendon b. ligament
   c. pectin d. fibrin
5. The bundle of muscle fibres is called
   a. Myofibrils b. fascicle
   c. sarcomere d. sarcoplasm
6. The pigment present in the muscle fibre to store oxygen is
   a. myoglobin b. troponin
   c. myosin d. actin